

conferences on molecular electronics in 1980s.

Although the progress of molecular electronics has not always been smooth, prospects for the future are bright.

Molecular electronics leading to bio/chemical computer

Molecular electronics is a revolutionary idea. In order to attain maximum miniaturisation, it is proposed that instead of transistor's 'on' and 'off' states, the characteristics of electrons may be used to implement 1's and 0's. For example, positive and negative spins can be used to implement 1's and 0's, respectively.

Molecular electronics is proposed to be based on a new organic material technology that may lead to bio or chemical computer. A new radical information processing system is being thought of where organic cells or bacteria will act as the basic element. Living organisms are made of organic compounds. As such, thinking function can be easily realised in such a system. As scaling will be done at biological level, very high-density circuits can be made.

Our brain comprises about 10^{11} neurons ranging in size (linear dimension) from 0.2 μm to about 100 μm , each with an average connectivity of 10^4 , giving a crude bit-count of 10^{11} to 10^{15} . An equivalent artificial brain may therefore be made of such a dense circuit. Enzymes and proteins are being studied. We should not forget the example of bacterial photo-reaction centre—a natural molecular device. Recent research to produce an analogous device has been successful through synthesis of single and complex molecules that release charge on photo-excitation.

Molecular electronics has moved from conjecture to experimental study and scientific development with the rapid development of a few liquid crystals, polymers, L-B films and NLO materials. It will prosper with ad-

vances in physics, chemistry, materials science, biology and engineering as our understanding of molecular materials both at microscopic and microscopic levels grows. A better understanding of natural system and processes will enhance the capability and potentiality of molecular electronics, particularly for radical new computational machines and engineering.

Quantum computing

While molecular electronics aims at attaining miniaturisation in line and/or beyond Moore's law, quantum computing applies quantum properties to information processing.

In quantum computing, the phase of the quantum wave function is proposed to represent and manipulate information. In classical physics, an electron can be at a high or low energy state. But in quantum physics, the electron is not just at high or low energy state but also at a superposition state. Thus a quantum bit contains more information than a classical bit. For example, each electron classically can be in either of two states, and thus a system of 100 hydrogen atoms will be in one of 2^{100} states. In quantum physics, the system at any position is at a state of superposition of all those 2^{100} states. It hints at the enormous resources of computing power.

The fundamental concept of quantum computing is quantum bits, referred to as qubit. A qubit is the abstract of an individual particle, say, electron, which has a probability of spinning one way or the other. It is in undecided state. This, in turn, states that qubit can be both 1 and 0 simultaneously. Qubits are superimposed quantum mechanical states that take over at atomic states. For example, qubits may be particles with '+' and '-' spin, or 'up' and 'down' spin. These may exist in logical states 1 and 0 as in classical computing and superposition of these classical logic states. Superposition refers to simultaneous existence of both 1 and 0 with probability of each state as a coefficient of existence. Thus

TABLE I
AND Function of
Classical Computer

Input 1 X	Input 2 Y	Output 1 Z
0	0	0
0	1	0
1	0	0
1	1	1

Qubit, $|\theta\rangle$, is a two-level quantum mechanical system with two basic states known as $|0\rangle$ (called ket 0) and $|1\rangle$ (called ket 1) such that each qubit may be represented as a combination of ket 0 and ket 1 (see Fig. 1). A qubit is actually the phase information of any quantum state.

A quantum computer is any device that processes and/or computes qubits as per instruction in the quantum algorithm. It exploits quantum mechanical phenomena to run algorithms as data is represented by the quantum properties of a given molecule or set of molecules.

Instead of making computations by retrieving data from a hard disk and processing with logic gates as in conventional computers, quantum computers process data by bombarding the information-containing molecule with short pulses of radiation. Each bombardment cycle represents an algorithmic operation on the data contained within the molecule. When the algorithm terminates, the quantum state of the molecule is measured—a process which itself biases the end result. This is due to the fundamentally uncertain nature of quantum mechanics. To tackle this difficulty, quantum computing algorithms are run multiple times and the weighted average of the output asymptotically approaches the correct answer. Because quantum mechanical phenomena are inherently probabilistic rather than deterministic, a well-defined answer in the first try is not possible.

Quantum computers possess certain capabilities that classical computers lack. Quantum computing allows